

Data Availability

The source code for the proposed SHED is publicly available at GitHub: <https://github.com/ydniuyongjie/SHED>.

Appendix A. MLLM-based Evaluation Prompt Templates

Appendix A.1. Object Similarity Evaluation Prompt

Background: We are evaluating the performance of a personalized image generation model. The goal is to determine how accurately the model preserves the identity of a specific object from a reference image when generating it in a new context.

- # Role: You are an impartial visual evaluator. You must strictly follow the scoring criteria. You are capable of distinguishing between stylistic changes and fundamental structural errors.
- # Task: Your task is to compare the Reference Object Image (I_{ref}) with the Generated Image (I_{gen}). You must quantitatively evaluate the similarity of the object itself (ignoring background and artistic style rendering).
- # Requirement:
 1. Analyze the images based on the following criteria and assign a normalized score from 0.0 (Worst) to 1.0 (Best) for each dimension:
 - Morphological Similarity (Weight: 0.4): Assess geometric vs. natural morphology. Is the shape consistent with the reference? (Score: 0.0 - 1.0)
 - Contour Similarity (Weight: 0.4): Focus on edges and outlines. Are straight lines/curves maintained? (Score: 0.0 - 1.0)
 - Compositional Structure (Weight: 0.2): Evaluate constituent relationships (e.g., specific parts like handles). Is the complexity level maintained? (Score: 0.0 - 1.0)
 2. Output Format: Provide the final evaluation strictly as a mathematical text formula as shown below.
 - Template: Object Similarity Score = $[\text{Morphological Score}] \times 0.4 + [\text{Contour Score}] \times 0.4 + [\text{Structure Score}] \times 0.2 = [\text{Final Weighted Score}]$
 - Reasoning: [One sentence explaining the deduction]

Appendix A.2. Style Similarity Evaluation Prompt

- # Background: We are evaluating a personalized image generation model. The goal is to determine how accurately the model transfers the artistic style from a reference image to a generated image, regardless of the object content.

Role: You are an impartial visual style evaluator. You must assess similarity only according to the defined criteria. You have deep expertise in analyzing brushwork, color theory, and composition techniques.

Task: Your task is to compare the Reference Style Image (I_{ref}) with the Generated Image (I_{gen}). You must quantitatively evaluate the similarity of the artistic style (ignoring the specific object content) based on four specific dimensions.

Requirement:

1. Analyze the images based on the following criteria and assign a normalized score from 0.0 (Worst) to 1.0 (Best) for each dimension:
 - Brushwork Characteristics (Weight: 0.3): Assess brushstroke style and texture. Specifically compare: Thick vs. Thin application; Bold vs. Delicate; Visible vs. Hidden brush strokes; and the orientation/morphology of brushstrokes. (Score: 0.0 - 1.0)
 - Color Application (Weight: 0.4): Evaluate the color scheme, saturation, and luminosity. specifically checking the Main color tone, Richness degree of colors, and Color contrast and harmony. (Score: 0.0 - 1.0)
 - Compositional Techniques (Weight: 0.1): Analyze composition methods and perspective. Look for Central vs. Dispersed composition, Symmetry vs. Asymmetry, and Perspective typography. (Score: 0.0 - 1.0)
 - Overall Effect (Weight: 0.2): Judge the holistic style and atmosphere. Assess Realistic vs. Abstract, Decorative vs. Expressive, Planar treatment vs. Volumetric rendering, and Kinetic vs. Static quality. (Score: 0.0 - 1.0)
2. Output Format: Provide the final evaluation strictly as a mathematical text formula as shown below.
 - Template: Style Similarity Score = [Brushwork Score] $\times$ 0.3 + [Color Score] $\times$ 0.4 + [Composition Score] $\times$ 0.1 + [Overall Effect Score] $\times$ 0.2 = [Final Weighted Score]
 - Reasoning: [One sentence explaining the deduction]